# METROPOLITAN CARDIOLOGY & DIAGNOSTIC INSTITUTE

CLINICAL CARDIOLOGY · ADVANCED NON-INVASIVE IMAGING · ACCREDITED LAB #CLIA-99214

452 Innovation Blvd, Medical Center Hub | Secure EHR Integration: HL7/FHIR Compliant

## COMPREHENSIVE CARDIOVASCULAR BIOMARKER REPORT

### PATIENT INFORMATION

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|---|---|---|---|
| Patient Name: | **Robert H. Vance** | Report Date: | **October 14, 2026** |
| Patient ID: | **CV-2026-8842** | Referring MD: | **Dr. Sarah Jenkins, MD, FACC** |
| Age / Gender: | **61 Yrs / Male** | Indication: | **Exertional Angina & Multi-Vessel CAD Screening** |

### SECTION 1: BLOOD BIOCHEMISTRY & SERUM LIPID PROFILE (FASTING)

| Biomarker Parameter | Result Value | Reference Interval | Clinical Interpretation |
|---|---|---|---|
| Total Cholesterol | **242 mg/dL** | < 200 mg/dL | **High (Borderline)** |
| LDL Cholesterol | **164 mg/dL** | < 100 mg/dL | **Elevated Risk (Critical)** |
| HDL Cholesterol | **34 mg/dL** | > 40 mg/dL | **Suboptimal / Low** |
| Triglycerides (TG) | **228 mg/dL** | < 150 mg/dL | **High** |
| Fasting Blood Sugar (FBS) | **148 mg/dL** | 70 - 99 mg/dL | **Impaired / Diabetic Range** |
| Serum Creatinine (Cr) | **1.15 mg/dL** | 0.70 - 1.30 mg/dL | **Normal Renal Clearance** |
| Blood Urea Nitrogen (BUN) | **19.0 mg/dL** | 7.0 - 20.0 mg/dL | **Normal** |
| Hemoglobin (Hb) | **14.2 g/dL** | 13.5 - 17.5 g/dL | **Normal** |

### SECTION 2: 12-LEAD RESTING ELECTROCARDIOGRAM (ECG) FINDINGS

- **Rhythm & Rate:** Normal Sinus Rhythm at 78 bpm. PR Interval: 164 ms | QRS Duration: 96 ms | QTc: 442 ms.
- **ST-Segment Changes:** Horizontal ST-Depression (1.4 mm) present in lateral precordial leads (V4, V5, V6).
- **T-Wave & Inversion:** Symmetric T-Wave Inversion observed in inferior leads (II, III, aVF). Pathological Q in Lead III.
- **Hypertrophy / Conduction:** Sokolow-Lyon voltage criteria positive for Left Ventricular Hypertrophy (LVH). No BBB.

### SECTION 3: 2D TRANSTHORACIC ECHOCARDIOGRAPHY (TTE) QUANTITATION

- **Ejection Fraction (LVEF):** **46% (Moderately Depressed; Biplane Simpson Method, normal >= 55%)**
- **Regional Wall Motion (RWMA):** Hypokinesis localized to anterior mid-cavity & apical septum (suggestive of LAD territory ischemia).
- **Valvular Heart Disease (VHD):** Mild Aortic Valve Sclerosis; Trace-to-Mild Mitral Regurgitation (Grade 1 VHD).
- **Diastolic Filling:** Grade I impaired relaxation pattern (E/A ratio 0.72).

### SYNTHESIS & CLINICAL IMPRESSION FOR ML TRIAGE:

Electrocardiographic lateral ST-depression and T-wave inversion combined with atherogenic dyslipidemia (LDL 164, TG 228), diabetes (FBS 148), and echo wall motion abnormality strongly point toward severe obstructive multi-vessel coronary artery disease. Immediate quantitative 3D coronary vascular triage and multi-vessel stenosis risk prediction (LAD, LCX, RCA) recommended.

**Verified by: Dr. Marcus Sterling, MD, FACC** **Accreditation: CAP / CLIA #05D2049182**